

Guided Comparative Assembly

Comparing construction methods through interactive physical assembly

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MISSION

We compare different statistics of the housing construction industry and display them through an interactive exhibit.

An interactive table where users physically assemble 2–3 scale models of the same object — each built with a different construction method.

As each piece is placed, production data is mapped directly onto the object: CO₂, labor

CONSTRUCTION METHODS

MASONRY

Brick by brick
Traditional

3D PRINT

Layer by layer
Automated

PREFAB

Panel by panel
Industrial

THE POINT

Instead of reading abstract statistics, you build the comparison yourself — piece by piece.

HOW IT WORKS

- 1 **Projector highlights**
next piece placement
- 2 **User places**
physical piece
- 3 **Camera confirms**
correct placement
- 4 **Data projected**
onto piece
- 5 **Repeat until**
model complete
- 6 **Next method**
or comparison

Data projected per piece:

● CO₂ consumption ● Labor hours ● Material origin ● Production method

Once all models are complete, a full comparison is projected across the table:
total CO₂, total labor, total cost — side by side.

Bonus: Arduino proximity sensor — user leans in → data zooms for detail.

The diagram illustrates a system architecture for a building assembly process. It is organized into three main horizontal layers: **INPUT** (top), **PROCESSING** (middle), and **OUTPUT** (bottom).

- INPUT Layer:** Contains the text "Human action = building." and "The act of assembly IS the input."
- PROCESSING Layer:** Contains several interconnected components:
 - Overhead USB webcam** (blue box) receives input from the "INPUT" layer and sends data to the "Arduino/ESP32 proximity sensor" and the "Processing" unit.
 - Arduino/ESP32 proximity sensor** (blue box) receives input from the "INPUT" layer and sends data to the "Processing" unit.
 - Processing** (blue box) receives data from the "Overhead USB webcam" and the "Arduino/ESP32 proximity sensor", and sends data to the "Output" unit.
 - Output** (blue box) receives data from the "Processing" unit and sends data to the "Human action = building." and "The act of assembly IS the input." text.
- OUTPUT Layer:** Contains the text "Human action = building." and "The act of assembly IS the input."

Arrows indicate the flow of information between these components.

Grasshopper + Anemone FSM

Anemone: FSM loop logic.
Firefly: webcam data stream
into Grasshopper.

Projection Feedback

TouchDesigner / HeavyM for
projection mapping layer.

```
graph LR; IDLE["IDLE  
Waiting for user  
Ambient pattern"] -- orange --> NEXT_MODEL["NEXT_MODEL  
Switch to  
next method"]; IDLE -- orange --> GUIDING["GUIDING  
Projector highlights  
next piece target"]; NEXT_MODEL -- yellow --> GUIDING; GUIDING -- yellow --> CHECKING["CHECKING  
Camera validates  
placement"]; CHECKING -- green --> CONFIRMED["CONFIRMED  
Data projected  
on piece"]; CONFIRMED -- red --> NEXT_PIECE["NEXT_PIECE  
Advance to  
next piece"]; NEXT_PIECE -- blue --> COMPARISON["COMPARISON  
Side-by-side  
final stats"]; MODEL_COMPLETE["MODEL_COMPLETE  
Method stats  
summary"] -.-> ERROR["ERROR  
Ghost guide  
shown"];
```

Tech Stack & Timeline

Tools, phases, and key reference

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TECH STACK

Rhino + Grasshopper	Main logic environment
Anemone (GH plugin)	FSM loop logic
Firefly (GH plugin)	Webcam/sensor → GH data stream
TouchDesigner / HeavyM	Projection mapping + visual output
Arduino / ESP32	Proximity sensor (lean in → zoom)
USB webcam (overhead)	Piece placement detection
Video projector (top-down)	Guided projection on table

KEY REFERENCE

Stefana Parascho — Cooperative Robotic Assembly (ETH Diss. 25839)

Assembly sequence as FSM: each intermediate state must be structurally valid before the next step is permitted. Directly maps to our approach

MORE REFERENCES

Gramazio Kohler — Augmented Bricklaying (ETH)
Fologram — Steampunk Pavilion (Tallinn 2019)
inFORM — MIT Tangible Media Group (2013)
EPFL SXL — Reuse-driven structural pavilions

PROJECT PHASES

